Hadoop local setup

U **You**

Original custom instructions no longer available

U **You** 5/5/2026, 7:22:05 PM

write command for confgruing hadoop on local change system files

starting service

installing if not

simply having simple file system like given by ls command

```
Directory: E:\DSBDA-main\Hadoop_Log_Analysis

Mode LastWriteTime Length Name
----------------------------------------------------------------
d----- 4/21/2026 12:05 PM output
-a---- 4/21/2026 12:05 PM 187505 Assign12_7323.png
-a---- 4/21/2026 12:05 PM 2837 LogAnalysis.jar
-a---- 4/21/2026 12:05 PM 1362 LogAnalysisDriver.class
-a---- 4/21/2026 12:05 PM 960 LogAnalysisDriver.java
-a---- 4/21/2026 12:05 PM 1710 LogMapper.class
-a---- 4/21/2026 12:05 PM 739 LogMapper.java
-a---- 4/21/2026 12:05 PM 1587 LogReducer.class
-a---- 4/21/2026 12:05 PM 527 LogReducer.java
-a---- 4/21/2026 12:05 PM 141 logs.txt
```

here input is logs.txt and output is saved in output folder

i know it should be deleted so if existed delete it , i have pre jar files but give commands for creating it

also dont be childish give me only useful response.

also this file sturcture is for windows but give me complete confriguration for linux only because it is my work machine.

AI **ChatGPT** 5/5/2026, 7:22:20 PM

Here's a **clean, Linux-only Hadoop setup + execution workflow** tailored to your use case (logs.txt → output folder, jar included + build commands).

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1) Install Hadoop (if not already)

```
# Install Java (required)
sudo apt update
sudo apt install openjdk-8-jdk -y

# Verify
java -version

# Download Hadoop
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz

# Extract
tar -xvzf hadoop-3.3.6.tar.gz
sudo mv hadoop-3.3.6 /usr/local/hadoop

# Set environment variables
nano ~/.bashrc
```

Add:

```
export HADOOP_HOME=/usr/local/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

```
source ~/.bashrc
```

---

2) Configure Hadoop (pseudo-distributed mode)

Edit files inside:

```
cd $HADOOP_HOME/etc/hadoop
```

---

core-site.xml

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

---

hdfs-site.xml

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
</configuration>
```

---

mapred-site.xml

```
cp mapred-site.xml.template mapred-site.xml
```

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
</configuration>
```

---

yarn-site.xml

```
<configuration>
 <property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
 </property>
</configuration>
```

---

3) Format HDFS (ONLY FIRST TIME)

```
hdfs namenode -format
```

---

4) Start Hadoop Services

```
start-dfs.sh
start-yarn.sh
```

Check:

```
jps
```

Expected:

```
NameNode
DataNode
ResourceManager
NodeManager
```

---

5) HDFS Basic Commands (like ls)

```
# Create directory
hdfs dfs -mkdir /user
hdfs dfs -mkdir /user/hadoop

# Upload file
hdfs dfs -put logs.txt /user/hadoop/

# List files (like ls)
hdfs dfs -ls /user/hadoop

# View file
hdfs dfs -cat /user/hadoop/logs.txt
```

---

6) Delete output folder if exists

```
hdfs dfs -rm -r /user/hadoop/output
```

---

7) Create JAR (from your Java files)

```
# Compile
javac -classpath `hadoop classpath` -d . LogAnalysisDriver.java LogMapper.java
LogReducer.java

# Create jar
jar -cvf LogAnalysis.jar *.class
```

---

8) Run MapReduce Job

```
hadoop jar LogAnalysis.jar LogAnalysisDriver /user/hadoop/logs.txt
```

```
/user/hadoop/output
```

---

9) Check Output

```
hdfs dfs -ls /user/hadoop/output
```

```
hdfs dfs -cat /user/hadoop/output/part-r-00000
```

---

10) Stop Services (when done)

```
stop-yarn.sh
stop-dfs.sh
```

Quick Flow Summary

- Install → Configure → Format
- Start services
- Upload `logs.txt`
- Delete old `output`
- Run JAR
- Check output

If you want, I can also give:

- ready-made `LogAnalysisDriver.java` template
- common errors + fixes (very useful in exams/practicals)